

The Chinese University of Hong Kong, Shenzhen

School of Data Science

MAT2041 · Assignment 2

Due: 17:00, March 3rd, 2026

Instructions. Please ensure that you complete this assignment independently. If your solution appears to be identical or highly similar to others or generated using AI, you may be required to meet with the TA to verify your understanding. Cases of academic dishonesty may result in penalties, including a zero score for the assignment or other actions by the disciplinary committee. If you have any questions or need assistance, feel free to reach out to the course staff for guidance. We encourage you to engage in the learning process with integrity and responsibility.

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Problem 1 (12 pts total, 4 pts each)

Find the values of k for which the system of equations with justification:

$$\begin{cases} kx + y + z = 1, \\ x + ky + z = 1, \\ x + y + kz = 1 \end{cases}$$

has (a) no solution, (b) exactly one solution, (c) infinitely many solutions.

Problem 2 (10 pts)

Verify the addition property for the transpose of matrices using

$$P = \begin{bmatrix} 4 & 0 & 3 \\ 5 & -1 & -2 \\ 1 & 2 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 5 & 1 & 1 \\ 3 & -4 & 0 \\ 2 & 1 & -3 \end{bmatrix}.$$

Problem 3 (12 pts total, 6 pts each)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 & 0 \\ 7 & 8 & 9 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ 1 & -1 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}.$$

Find AB using two ways: (a) direct multiplication, (b) block multiplication.

Problem 4 (12 pts total, 4 pts each)

Let A, B, C be $n \times n$ matrices. The commutator is defined as

$$[A, B] = AB - BA.$$

Find X and Y (justify your assertions):

- (a) $[A, B]C + B[A, C] = [X, Y]$
 (b) $[A, C] + [B, C] = [X, Y]$
 (c) $[[A, B], C] + [[B, C], A] + [[C, A], B] = X$

Problem 5 (12 pts total, 6 pts each)

- (a) Suppose $\mathbf{v} \in \mathbb{R}^{m \times 1}$ and $A \in \mathbb{R}^{m \times n}$. The product $\mathbf{v}^\top A$ is a $1 \times n$ vector. Express $\mathbf{v}^\top A$ in three forms:
- a formulation relying on the definition of matrix multiplication,
 - a formulation relying on the column partition of A ,
 - a formulation relying on the row partition of A .
- (b) Let us denote the columns of A by $\mathbf{a}_1, \dots, \mathbf{a}_n$ so that

$$A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n].$$

Express $A^\top$, $AA^\top$ and $A^\top A$ using $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Problem 6 (12 pts, 6 pts each)

- (a) Suppose U and R are $n \times n$ upper triangular matrices and $T = UR$. Show that T is also upper triangular and that $t_{ii} = u_{ii}r_{ii}$ for all $i = 1, \dots, n$.
- (b) Suppose A is an upper triangular $n \times n$ matrix and B is an arbitrary $n \times n$ matrix. Denote $C = AB$. Alice conjectures that $c_{ii} = a_{ii}b_{ii}$ for all $i = 1, \dots, n$. Is this conjecture true? If yes, prove it; if no, provide a counter-example.

Problem 7 (15 pts total, 3 pts each)

Judge whether each statement is true or false. If true, provide a brief explanation why it is true. If false, provide a counter-example (i.e., a specific example that the statement does not hold).

- (a) If a square matrix A satisfies $A^2 = 0$, then A is a zero matrix.
- (b) Suppose 2×2 square matrices A, B, C satisfy $AB = AC$, then we must have $B = C$.
- (c) For any $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times k}$ where $k \geq 3$, if the first column and the third column of B are the same, then the first column and the third column of AB are the same.
- (d) For any $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times k}$ where $n \geq 3$, $m \geq 3$, if the first row and the third row of matrix B are the same, then the first row and the third row of AB are the same.
- (e) There exists a matrix $A \in \mathbb{R}^{m \times n}$ such that for any matrix $B \in \mathbb{R}^{n \times k}$, every row of AB is the same as the first row of B .

Problem 8 (15 pts)

Find a counter-example or prove the following statement:

There exists a matrix A such that

$$AM = M^\top$$

for any $n \times n$ matrix M .